# MILITARY SPECIFICATION

## CATALYST, BURN-RATE

## 2, 2- BIS (ETHYLFERROCENYL) PROPANE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for a burn-rate catalyst for use in the manufacture of solid rocket propellant, referred to herein as the catalyst.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| BB-N-411 | Nitrogen, Technical. |

MILITARY

| | |
|---|---|
| MIL-A-18455 | Argon, Technical. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-1218 | ACS Chemicals. |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARDS

American Society for Testing and Materials (ASTM)

| | |
|---|---|
| ASTM D-445 | Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity), Test For. |

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS.

3.1 Chemical and physical properties. Chemical and physical properties of the catalyst shall conform to Table I.

TABLE I. Chemical and physical properties.

| Property | Min | Max |
|---|---|---|
| Viscosity at 25°C, centistokes | --- | 2,200 |
| Iron content, percent | 23.3 | 24.3 |
| Moisture, percent | --- | 0.10 |
| Acidity, milliequivalents/gram | --- | 0.025 |
| Insolubility, percent | --- | 0.1 |
| Evaporation loss, percent | --- | 4.0 |
| Assay (by titration), percent | 97.5 | --- |



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3.2 Stability. When packaged in accordance with 5.1.1 and stored at 27 ±11 degrees Celsius (°C), the catalyst shall have a storage life of 12 months from date of delivery to the procuring activity. The shelf life may be extended for 6-month intervals after reacceptance testing for conformance to viscosity and moisture limits.

3.3 Toxic products and safety. Safety regulations and guidelines applicable to the use of the catalyst should be complied with to preclude personal injury and damage to equipment and facilities.

3.4 Workmanship. Workmanship shall be such that the catalyst is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2).

4.2 Sampling. The lot shall be sampled in accordance with Table II.

TABLE II. Sampling plan.

| Number of containers in lot | Number of containers sampled (primary sample) | Number of composite samples |
|---|---|---|
| 100 or more | 10% (nearest whole number) | 5 |
| 51 - 99 | 10 | 4 |
| 11 - 50 | 10 | 3 |
| 1 - 10 | ALL | 2 |



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4.2.1 Primary samples. Physical properties tests shall be run on each primary sample (see Table II). The material container may be sampled by use of a clean glass tube, rod, or pipet. If the container is small enough to be handled safely, the sample may be obtained by pouring. The smallest sample size possible that is consistent with test requirements shall be taken. The minimum sample size shall be two ounces. Glass containers shall be used for all liquid samples. Each sample shall be labeled with date, lot number, and manufacturer's container identification number. Failure of any primary sample to pass all of the physical-properties tests herein shall result in rejection of the lot represented.

4.2.2 Composite samples. Chemical properties tests shall be run on each composite sample. Divide the primary samples equally into the number of composites shown in Table II. Blend each composite thoroughly by manipulation of the container. Label each composite with Roman numerals, also include date, lot numbers, and manufacturer's container identification numbers. The remainder of the primary samples shall be retained pending acceptance or rejection of the lot. Failure of any composite sample to pass all of the chemical-properties tests herein shall result in rejection of the lot represented.

4.3 Quality conformance inspections. Quality conformance tests and inspections shall consist of the following for each lot of catalyst material (see 6.3.1):

a. Tests of Table I properties (see 4.4).

b. Inspection of filled containers (see 4.6.1).

c. Visual inspection (see 4.6.2).

4.4 Test methods. Tests shall be performed using apparatus, reagents, and procedures specified herein. The use of alternate apparatus, reagents, or procedures shall require prior written approval of the procuring activity. All American Chemical Society (ACS) reagents shall conform to MIL-STD-1218.

4.4.1 Viscosity.

4.4.1.1 Apparatus.

a. Cannon-Fenske viscometer as specified by ASTM D 445, or equal.

b. Water bath with thermostatic control to ±0.05°C in the range from 20 to 50°C.



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4.4.1.2 Reagents.

a. Standard viscosity oil, type as specified by ASTM D 445, or equal.

4.4.1.3 Determination of viscosity. Position the appropriate viscometer in a 25°C constant temperature water bath. Align the measuring section to vertical. Using a pipet or syringe, place 7 to 8 milliliters (ml) of sample in the viscometer reservoir. Allow the tube to remain in the constant temperature water bath for 15 minutes. Use a rubber bulb to adjust the head level of the sample to approximately 5 millimeters (mm) above the upper timing line. Release the vacuum and measure the time required for the meniscus to pass from the first to the second timing mark. Calculate the average of at least two readings on the sample in the tube.

Calculation:

Viscosity, 25°C, centistokes = T x C

Where: T = average efflux time, seconds

C = calibration constant provided with the instrument or obtained with standard oil at 25°C

Report the viscosity to the nearest one centistoke.

4.4.2 Iron content.

4.4.2.1 Apparatus.

a. Funnel with Tygon, or equal, tubing connected to water aspirator.

b. Meeker burner, or equal.

4.4.2.2 Reagents.

a. Hydrochloric acid, concentrated, ACS reagent.

b. Nitric acid, concentrated, ASC reagent.

c. Sulfuric acid, concentrated, ACS reagent.

d. Perchloric acid, 70 percent, ACS reagent.

e. Stannous chloride solution, 5 percent (25 grams (gm)/500 ml of 10 percent hydrochloric acid plus a few granules of tin).

f. Mercuric chloride solution, saturated.



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g. Phosphoric acid, 85 percent, ACS reagent.

h. Sodium diphenylamine sulfonate indicator solution, 0.4 percent.

i. Potassium dichromate, primary standard, 0.1 Normal (N).

4.4.2.3 Determination of iron content. Accurately weigh 0.50 gram of sample into a 250-ml Erlenmeyer flask. Add 10 ml of distilled water and 15 ml of concentrated nitric acid.

CAUTION

Add water prior to acid or sample ignition will occur.

Swirl to complete dissolution. Swirl contents carefully and constantly while adding 5 ml of concentrated sulfuric acid. Add 1 ml of concentrated, 70 percent perchloric acid and several boiling beads. Invert funnel over the flask and connect the funnel to a water aspirator with Tygon, or equal, tubing. Heat the sample slowly over a burner and increase the heat to digest at dense white fumes for 2 to 3 minutes. Add 1 ml of concentrated nitric acid and five drops of perchloric acid and heat to fumes again. Fume for 3 to 5 minutes. Cool to safe temperature, add 25 ml of distilled water and reheat until precipitate is completely dissolved. While solution is hot, add 10 ml of concentrated hydrocholoric acid and then 5 percent stannous chloride solution, dropwise, until one drop changes the color no more. (The solution may be light pale yellow or colorless.) Add one to two drops in excess. While stirring the solution, add (at one time) 10 ml of saturated mercuric chloride solution and let stand 10 minutes. Add 5 ml of 85 percent phosphoric acid, 100 ml of distilled water, three drops of sodium diphenylamine sulfonate indicator solution, and titrate with standard 0.1 N potassium dichromate solution to a violet end point.

Calculation:

$$\text{Iron, percent} = \frac{V \times N \times 5.585}{W}$$

Where: V = volume of standard potassium dichromate, ml

N = normality of standard potassium dichromate

W = weight of sample, gm

Report the iron content to the nearest 0.1 percent.

4.4.3 Moisture.

4.4.3.1 Apparatus.

a. Beckman "Aquameter", Model KF-4, or equal.



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4.4.3.2 Reagents.

a. Karl Fischer (KF) reagent, commercial strength.

b. KF reagent, stabilized, water equivalent of 2.0 to 3.0 milligrams/milliliter (mg/ml).

c. Pyridine-chloroform, ACS reagent, 1:1 by volume.

4.4.3.3 Determination of moisture. Add 10 gm of sample to a reaction beaker containing 60 ml of neutral pyridine-chloroform solution. The stirrer should be off while adding the sample. Turn on the stirrer to the same position used for neutralization. Titrate automatically, by pressing the titrate button to a 30-second end point.

Calculation:

$$\text{Moisture, percent} = \frac{V \times E}{10 \times W}$$

Where: V = volume of standard KF reagent, ml

E = water equivalent of KF reagent, mg/ml

W = weight of sample, gm

Report the moisture to the nearest 0.01 percent.

4.4.4 Acidity.

4.4.4.1 Apparatus.

a. pH indicator, Leeds and Northrup, number 7664 equipped with glass electrode standard 1199-30 and calomel electrode standard 1199-31, or equal.

b. Magnetic stirrer, variable speed, with Teflon coated stirring bar, or equal.

4.4.4.2 Reagents.

a. Methanol, ACS.

b. Standard sodium hydroxide solution, 0.1 N.

4.4.4.3 Determination of acidity. Accurately weigh 2.5 gm of sample into a 250-ml beaker. Add 50 ml of methanol, 50 ml of distilled water, and stir to disperse. Titrate potentiometrically with standard 0.1 N sodium hydroxide solution. Determine a reagent blank in like manner. Determine the end point graphically by plotting the volume of titrant versus millivolts, taking the midpoint of the straight section of the curve as the end point.



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Calculation:

$$\text{Acidity, milliequivalents/gram of sample} = \frac{(V - B)\,N}{W}$$

Where: V = volume of standard sodium hydroxide for sample, ml

B = volume of standard sodium hydroxide for blank, ml

N = normality of standard sodium hydroxide

W = weight of sample, gm

Report acidity to the nearest 0.001 milliequivalents per gram sample.

4.4.5 Insolubility.

4.4.5.1 Apparatus. Filter paper, Whatman number 1, or equal.

4.4.5.2 Reagents. Chloroform, ACS reagent.

4.4.5.3 Determination of insolubility. Accurately weigh 10 gm of sample into a 250-ml beaker and dissolve in 100 ml of chloroform. Filter the dissolved sample through an accurately weighed number 1 Whatman filter paper, or equal. Wash paper and contents with chloroform to remove any soluble sample. Dry filter paper and contents in a 105°C oven for 1 hour, cool in a desiccator, and reweigh.

Calculation:

$$\text{Insoluble material, percent} = \frac{(A - B)\,100}{W}$$

Where: A = weight of filter paper and residue, gm

B = weight of filter paper, gm

W = weight of sample, gm

Report the insoluble material to the nearest 0.1 percent.

4.4.6 Evaporation loss.

4.4.6.1 Apparatus.



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a. Aluminum weighing pans, disposable,

b. Vacuum oven.

4.4.6.2 Determination of evaporation. Accurately weigh 10 gm of sample into a disposable aluminum pan and place in a vacuum oven at 80°C and 15 mm mercury for 22 hours. Remove from oven, cool in a desiccator, and reweigh.

Calculation:

$$\text{Evaporation loss, percent} = \frac{(W - A)\ 100}{W}$$

Where: A = weight of sample after heating, gm

W = weight of sample before heating, gm

Report the evaporation loss to the nearest 0.1 percent.

4.4.7 Assay.

4.4.7.1 Apparatus.

a. Titrator, Methrohm E-436 or E-536, or equal, with combination platinum electrodes and 10 ml buret.

b. Beakers, 10 ml and 100 ml.

c. Graduated cylinder, 100 ml.

d. Syringe, tuberculin, 1 ml.

e. Glass frit filter, medium porosity.

4.4.7.2 Reagents.

a. Cupric perchlorate, 0.1 N standardized (see 4.4.7.3), reagent grade.

b. Tetrahydrofuran (THF).

c. Acetonitrile, reagent grade.

d. Ferrocene, reagent grade.

f. Perchloric acid, 1 N in water.



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4.4.7.3 Preparation of standard 0.1 N cupric perchlorate.

a. Weigh 37.0 gm of cupric perchlorate into a 1-liter bottle and dilute to the mark with acetonitrile. Shake to dissolve.

b. Filter through a medium porosity sintered glass filter. Mix well and store in a bottle with a screw cap polycone liner. This solution must be filtered prior to each standardization if any turbidity or solids are noted. Examine carefully.

c. Set up the titrator according to the manufacturer's instructions. Typical conditions are:

| | E-436 | E-536 |
|---|---|---|
| Speed: | 3 | 10 min/100% volume |
| Measuring range: | 1.5 volt (V) | 1.5 V |
| Mode: | millivolt (mV) | mV |

d. Accurately weigh 0.100 gm of ferrocene into a 100 ml beaker and dissolve in 60 ml of acetonitrile.

e. Add 0.2 ml of 1 N $HClO_4$ to the dissolved ferrocene and immediately titrate potentiometrically against the cupric perchlorate solution using a combination platinum electrode. If a 20 ml buret is used, double all quantities.

NOTE: Do not add the perchloric acid until the titration is set to begin.

f. Determine the volume of cupric perchlorate solution used from the titration curve.

g. Fill out the standardization label calling for daily restandardization.

Calculation:

$$N\ Cu(ClO_4)_2 = \frac{gm\ ferrocene \times 1000}{ml\ Cu(ClO_4)_2 \times 186.03}$$

4.4.7.4 Procedure.

a. Accurately weigh approximately 0.18 to 0.20 gm of catalyst into a 100 ml beaker.

NOTE: Double the sample size if using a 20 ml buret.

b. Add 25 ml of THF and stir to dissolve. Add 40 ml of acetonitrile and stir.



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c. Set up the titrator according to the manufacturer's instructions. Typical conditions for the Metrohm titrator are:

| | E-436 | E-536 |
|---|---|---|
| Speed: | 3 | 10 min/100% volume |
| Measuring range: | 1.5 V | 1.5 V |
| Mode: | mV | mV |

d. Titrate the sample solution against 0.1 N cupric perchlorate using a combination platinum electrode.

NOTE: These solutions are air sensitive, therefore titrate as soon as the sample is in the solution. Do not whip air into the solution while stirring.

e. Locate the equivalent point, determine a blank and calculate as follows:

$$\text{Percent catalyst} = \frac{(\text{ml} \times \text{N})\ \text{Cu}\ (\text{ClO}_4)_2 \times 46.825}{\text{gm sample} \times 2}$$

Report percent catalyst to the nearest 0.1 percent.

4.5 Stability. The material shall be tested in accordance with 4.4.1 and 4.4.3 after 12 months storage and at 6-month intervals thereafter to extend the shelf life for additional 6-month intervals (see 3.2).

4.6 Examinations.

4.6.1 Inspection of filled containers. All filled containers shall be inspected prior to shipment or use for accuracy of markings and for defects in containers and closures. All defective containers and closures shall be repaired or replaced, and contents therein shall be reinspected prior to shipment or use.

4.6.2 Visual inspection. All samples shall be visually inspected to determine conformance to the requirements of 3.4.

4.7 Records. Certification and test data shall be supplied as required by the procuring activity (see 6.2.2).

5. PACKAGING.

5.1 Preservation-packaging and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packaging and packing of the material shall be in accordance with commercial practice to insure carrier acceptance and shall be of such construction and materials that the contents will be adequately protected against loss or contamination.



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5.1.1 Special packaging. Each container shall be purged with argon conforming to MIL-A-18455 or nitrogen conforming to BB-N-411, Type I, Class 1, Grade A for a minimum of 6 minutes at a flow rate of 10 standard cubic feet per hour, minimum. The container shall then be sealed and marked before shipment (see 5.2).

5.1.2 Repackaging. When required, the material shall be repackaged into clean, dry containers. Containers that are opened for material removal shall be purged with argon or nitrogen, sealed and marked in accordance with 5.1.1.

5.2 Marking for shipment. Unless otherwise specified in the contract or purchase order (see 6.2.1), each shipping container shall be marked in accordance with the requirements of MIL-STD-129. Container marking shall include the following:

a. The supplier's lot number.

b. Procuring activity purchase order number.

c. Container identification number (applied in numerical sequence as the containers are filled).

d. Manufacturer's Code Ident.

e. Date of manufacture.

f. Net and tare weight of the container.

g. Material identification.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The intended use of the material described herein is as a burn-rate catalyst in propellant for rocket motors.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities if different than 4.1.

c. Special packaging, packing, or shipping requirements, if applicable (see Section 5).



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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

| Paragraph | Data Requirement | Applicable DID |
|---|---|---|
| 4.7 | Certification | UDI-A-23264B |
| | Test Data | DI-T-4024 |

(Copies of DIDs required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Lot. At place of manufacture, a lot consists of one batch (see 6.3.2) or a uniform blend of two or more batches. At place of delivery, a lot consists of material from one supplier's lot received in a single shipment. Partial shipments may be considered as a single shipment by the procuring activity.

6.3.2 Batch. A batch consists of material made as one unit in an unchanged manufacturing process.

6.4 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is marketed by Arapahoe Chemicals, Inc, 2075 N. 55th St, Boulder, CO 80301. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification or as any limitation of additional potential sources of supply.

Preparing activity:
Navy - AS

(Project 6810-NB18)



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-85493(AS), CATALYST, BURN-RATE 2,2-BIS (ETHYLFERROCENYL) PROPANE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

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